



Diagramming Working Field Theories for Design in the HCI Classroom

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ABSTRACT

HCI has historically provided little support for moving from field-work insights or theories to design outcomes. Having witnessed many students struggle and then justify their designs with a form of marketing hype, we developed a supporting approach of “field theories”. A field theory is a working theory about salient interactions in a particular domain and sensitizing concepts in order to frame design investigations. It is presented visually in a field theory diagram to support succinct communication and critique. Studying use of design prototypes that have been informed by a field theory helps to reflect upon and refine the theory. In this paper we present examples from our HCI classes and reflections based on interviews with students. We discuss how field theories offer an orientation in the spirit of a ‘bricoleur’ who harnesses elements of theory and practice to produce deeper understandings and more fitting outcomes for the task at hand.

CCS CONCEPTS

• **Human-centered computing** → HCI theory, concepts and models.

KEYWORDS

HCI education, fieldwork, sensitizing concept, visual thinking, design research

ACM Reference Format:

Bernd Ploderer, Jennyfer Lawrence Taylor, Diego Muñoz, Filip Bircanin, and Margot Brereton. 2021. Diagramming Working Field Theories for Design in the HCI Classroom. In *CHI Conference on Human Factors in Computing Systems (CHI '21)*, May 08–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 14 pages. <https://doi.org/10.1145/3411764.3445414>

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CHI '21, May 08–13, 2021, Yokohama, Japan

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ACM ISBN 978-1-4503-8096-6/21/05...\$15.00

<https://doi.org/10.1145/3411764.3445414>

1 INTRODUCTION

A core endeavour of the field of Human Computer Interaction research is to imagine and design new forms of interaction. Designers and researchers fashion our existing world and practices into new kinds of interaction by studying the existing and reconfiguring it into new ways of doing and seeing [49]. However, there is no simple recipe or linear path to get from the existing to the new.

Design, as described by Schön, is a reflective conversation with materials of a design situation [43; 45]. In the case of interaction design, these materials are many and varied, ranging from physical materials, new technologies, old technologies, and software to conceptual materials such as field research findings, identified needs, co-design discussions and workshops, interaction concepts and theories, fiction, design practice, and emerging societal problems, to name a few. In the reflective conversation, as the design evolves, so too does the understanding of the problem-opportunity space. This is important to note because it has been common to think of preliminary fieldwork and user-centred approaches as somehow feeding in at the start of the design process with the results of these processes forming a set of requirements for design.

In this paper we describe an approach, which we call “working field theories”, that aims to keep both the design context, sensitizing concepts and the design path in play. Field theories capture working knowledge about a particular field of design inquiry and sensitizing concepts that highlight opportunities for the design of interactive technology. We developed field theories especially for use in teaching, because we often see a discontinuity between studies of the design context, HCI theories and the design proposals of our students. Leaps to designs are made that seem to miss out on interesting subtleties and nuances of both interactions in the field and less obvious possibilities of new technologies. Moreover, theoretical concepts are often overlooked because cannot be easily translated to the problem at hand.

We have therefore sought an approach in teaching that entangles both studies of the design context, sensitizing concepts and design proposal making. Anderson cautioned that the purpose of ethnography has often been mistaken by designers as a form of data collection, but, as Anderson argues, the deeper understandings gleaned from ethnographic work can “enable designers to question the assumptions embedded in the conventional problem-solution

design framework” and serve as a powerful means to “open up the play of possibilities for design” [1]. We have therefore sought ways to keep the problem-opportunity space more explicitly in dialogue with the process of design.

Our concerns relate to both interaction design practice and research through design. There is a sense in which all interaction design is a form of research as it produces new knowledge. The resulting knowledge may take the form of exemplar designs in response to a problem or context, possibly accompanied by a rationale, as is typical in interaction design practice. In addition, knowledge may be further synthesised by studying and reflecting on use of the new design in different situations and conceptualising this in relation to prior research, as is more typical when doing research through design.

While we appreciate traditional design school based approaches centred on the sketching and artefact prototyping [13; 23], our own practice has a much more hybrid flavor based in co-design, ethnographic data, sensitizing concepts, as well as in artefact prototyping and long-term studies of use in-situ. Moreover we teach students who come from a variety of different disciplines about interaction design and engage students in research through design. Typically they come from either technical disciplines (computer science, information systems, games design), or from various social science disciplines (psychology, anthropology), but relatively few are from design school backgrounds. As such we approach both undergraduate teaching and doctoral research through design work, through a combination of understanding HCI theories, undertaking fieldwork and envisioning and prototyping designs. These activities are mutually informing.

Although techniques derived from various design practices are often taught to HCI students, such as brainstorming, personas, scenarios, journey maps, emotion maps, how-why diagrams, cultural probes, synectics, etc., it is possible for students to approach these in a task oriented fashion, and to lack an appreciation of the gestalt of a design idea [44]. In this case the whole, in terms of design insight may be no more than the sum of some uninspired parts. And while many HCI research projects artfully transition from fieldwork inspirations to interesting designs, the process of doing so often remains implicit. Given the interdisciplinary nature of HCI research, involving fields ranging from software engineering to psychology to design, and the variety of materials, concepts and fieldwork insights in play, we believe there is a need to provide more explicit support for their integration.

While there is no clear recipe for moving between these sources of inspiration and design [17; 22], we present field theories, which make explicit a working understanding of salient interactions in a particular domain and sensitizing concepts in order to frame design investigations, so that these understandings can be recognised, synthesised and challenged.

2 RELATED WORK

Research on HCI education highlights that our concerns are not unique. In a survey of HCI educators, Wilcox et al. [54] identified a divide between formal iterative design that is based on traditions of user-centred design and empirical research evidence, and creative design that is driven by ideation and design expertise

in a studio-based tradition. Oguamanam et al. [38] found that addressing this divide by introducing a studio-based approach into HCI classes for computer science students leads to a cultural clash. Hence, with the field theory we seek to offer a blended approach for students with a technical background develop a sensitivity for empirical research and creative design.

2.1 Knowledge from the Field

In terms of empirical research, field theories have been inspired by user-centred design approaches like rapid ethnography [34] and contextual inquiry [3]. These approaches employ ethnographic techniques of spending time with people through observations and interviews, but with a limited scope and timeframe. They reveal insights into current practices and experiences in a particular setting, which can shape requirements for design.

Design methods like cultural probes [24] and toolkits [42] offer additional insights. Cultural probes invite people to reflect on their experience in playful ways. Whilst they were conceived to provide “inspirational data” [24], many HCI researchers now incorporate cultural probes into their interviews to generate a better understanding of the design context [7; 48]. Toolkits used in co-design workshops tap into the creativity of participants, e.g., by encouraging participants to make a collage, map, or a rough prototype. These artefacts aid participants in recalling past experiences or in imagining future experiences [41; 42].

Analytically, HCI researchers often employ inductive qualitative data methods, like grounded theory [11] and thematic analysis [9]. Whilst these methods are different in terms of rigour and theoretical foundation, researchers employing such methods generally construct descriptive insight by creating thick descriptions or by abstracting (coding and identifying themes) observations and conversations with study participants. Grounded theory asks researchers to constantly compare evidence between different participants and settings, to recruit participants purposefully to test or refine preliminary knowledge, and to reflect on one’s interpretations through analytic memos. Sensitizing concepts [8] from existing theory can be incorporated into a grounded theory to create conceptual insight. Rigour is important in grounded theory and typically argued for in terms of the fit of the theory, its usefulness and applicability [11].

Diagrams play an important role to illustrate and label important insights about the design context and to communicate these insights in an accessible manner to participants and stakeholders. In UX design practice, common diagrams include personas [39] customer journey maps [27], and rich pictures [35]. Such UX deliverables are not typically considered theories, i.e., they are not generated with academic qualities in mind such as applicability or defensibility. However, they still offer a rhetorical quality because they help researchers to describe things to themselves as well as to others [28]. Diagrams are also common to academic research, e.g., through maps and scenes in ethnographic field notes [17], thematic maps in the analysis process [9], and experience models [6] that represent observational data and patterns to inspire and guide the design of interactive technologies.

There is no shortage of methods to create and visualise knowledge about the field of design inquiry. What is missing from these methods, we contend, is an integration and testing through

creative design. We acknowledge that the above-mentioned methods involve testing. For example, grounded theories are tested and iterated through cycles of empirical research and analysis until a point of saturation is reached [11]. Similarly, user-centred design employs cycles of testing and iterating prototypes and requirements [46]. However, these tests are based on a priori assessments of good design [54], rather than their ability to inspire designs that are creative and fitting to a design situation.

2.2 Knowledge from Design

Design-oriented research in HCI typically creates new knowledge from designing innovative interactions and from evaluating them empirically through use studies [30]. Knowledge from design-oriented research is often referred to as intermediate-level knowledge, because it occupies a middle ground between concrete instances of design and generalizable theories [30]. Intermediate-level knowledge is generative, that is knowledge that enables designers to create or invent something new.

A number of strategies have been proposed to document and visualise intermediate-level knowledge. For example, annotated portfolios [13; 23] make explicit a designer's concerns and judgments across a collection of particular designs. According to Gaver [23], the role of theory in annotated portfolios is *not* to generalize from particulars of design, its role is simply to provide inspiration and annotation. Strong concepts are interaction design knowledge (interface elements and interaction practices) abstracted from particular instances of design [30]. These abstractions provide opportunities for other designers to generate new instances of design. Through a process of vertical grounding, design researchers are asked to strengthen connections between a strong concept and other specific instances, as well as to broader theories. While strong concepts emerge from an artefact, bridging concepts [14] are inspired by exemplars of design and theoretical foundations. From these foundations, untried design opportunities are articulated and links to practical design exemplars are made.

These strategies have in common that they take a particular design as a starting point. Knowledge is abstracted from a number of design instances, which in turn enables designers to create or invent something new. What is missing from such intermediate-level knowledge, are the insights about the field of inquiry that have framed and inspired a design in the first place. The lines of inquiry proposed by Desjardins and Key [16] account for the stories and insights by which designers arrive at a particular design. However, this approach also starts from the design, and then looks back at how the designers have arrived at a certain design and what other outcomes may have been possible. What is missing, we argue, is an intermediate-level knowledge where fieldwork insights, theoretical concepts, and artefact design evolve in parallel and shape each other.

3 FIELD THEORIES

3.1 Introducing Field Theories

A field theory is a working theory that organises key characteristics about a particular field of design inquiry that highlights opportunities for the design of interactive technology. A theory, in general, is a system of interconnected ideas that condenses and organises

knowledge [36]. The field, from an HCI perspective, comprises different elements like people, interactions, practices, settings, technology etc. that come together in a particular situation [40]. Field theories are hybrids that synthesise insights into the field from field research (research *for* design) with insights from designing and testing prototypes in the field (research *through* design). They provide intermediate-level knowledge from particular instances of interaction design research [30]. However, unlike other forms of intermediate-level knowledge [23; 30], field theories are not produced (or abstracted from) after an artefact has been designed. Instead, a field theory constitutes the very knowledge that frames and enables the design of artefacts, and trialling the artefact is a means to test and refine the field theory.

Field theories are expressed as a diagram so that others can quickly get an understanding of key interactions and experiences that frame the design of new artefacts. The ultimate aim is for the diagram to work as a standalone summary of key insights. Figure 1 shows a diagram created by students based on field research with physiotherapy patients. We use this example in class because it clearly illustrates key components of a field theory diagram: the diagram usually fills one A4 page to give an overview of important problems and design opportunities, without overwhelming the reader. It includes a title and succinct definition to frame the field theory. This is followed by visual representation of empirical insights and related theoretical concepts that highlight problems and opportunities for designing new technologies. Whilst this example is presented as a journey from the perspective of a patient, we highlight to students that there is no fixed diagram structure for field theory diagrams. Field theories are deliberately flexible in the way they can be visualised to fit a specific field of inquiry.

Field theories are *working* theories for design in a particular context: a humble theory [22; 29] from a small set of participants that presents working knowledge for the design and trialling of artefacts, without making claims for substantive knowledge as in a generalizable theory. In this sense, field theories are *generative* as defined by Höök and Löwgren [30], because they present requirements that play a direct role for the design of technology or for further investigation by revealing interesting gaps or perspectives to be pursued. Furthermore, field theories are *rhetorical* as discussed by Halverson [28], because they help us talk about the world by labelling and describing important insights into the field.

3.2 Classroom Context and Positionality

Our approach to field theories has evolved through our interaction design course for computer science and engineering undergraduate students. This is an elective unit, taken by approximately 40 students each year. The course builds on prior courses on design thinking and user research. During the 13 weeks of the course, students work on a project that follows a typical design process, from understanding the problem domain, through to defining the problem, ideation, and prototype trials. Weekly workshops provide background information through mini-lectures, as well as ongoing support and feedback on the students' project work.

We have collaboratively taught field theories since 2016. Author 1 is an early-career academic in HCI with a background in informatics, with an affinity for traditional qualitative research methods like

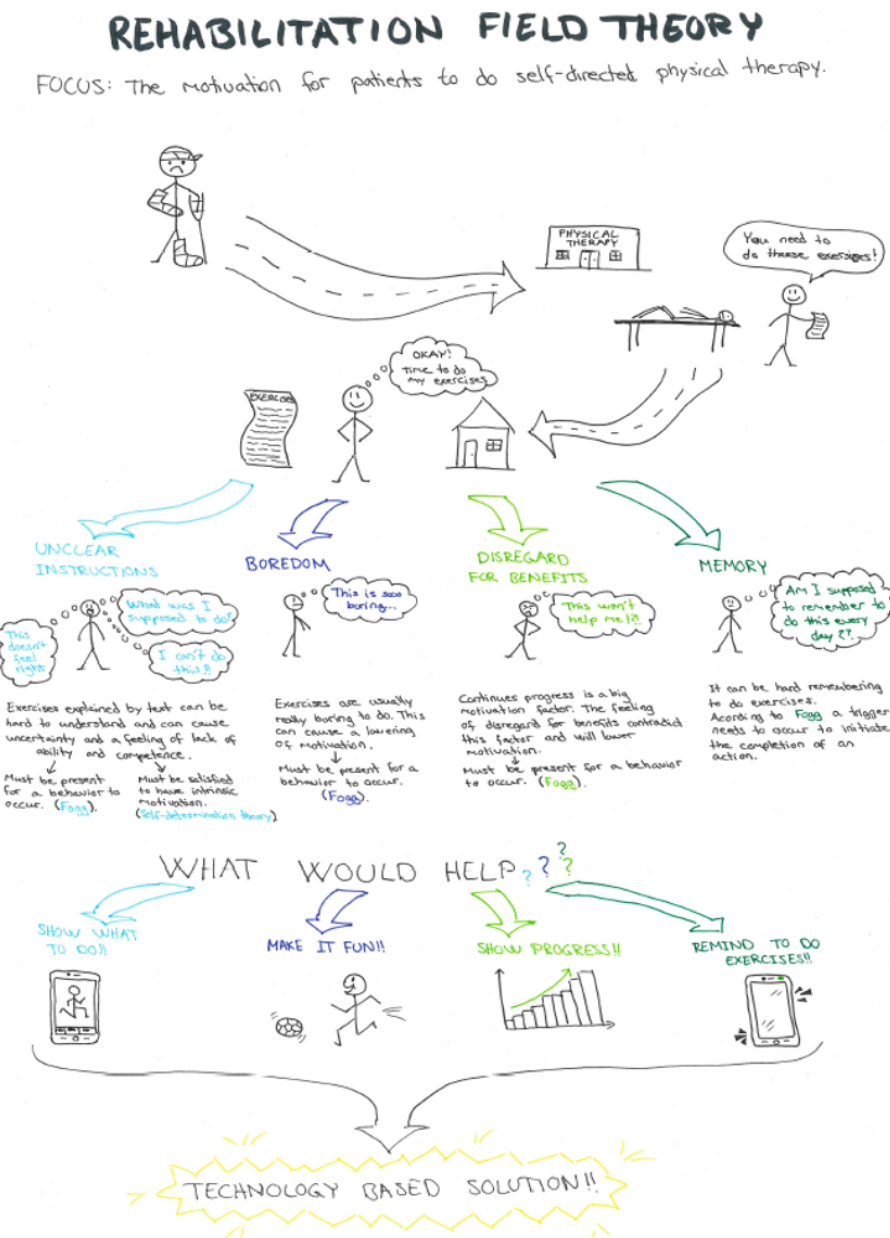


Figure 1: Field theory diagram and accompanying VR design. The diagram provides insight into a patient's experience with therapy instructions. Guided by field research and theories of motivation [19], it highlights four design opportunities: clear instructions, addressing boredom, highlighting benefits, and providing triggers to remember exercise. Students designed a VR prototype to test the assertion that clear instructions would indeed enhance the experience of patients, which allowed them to refine the field theory.

grounded theory [11]. Author 2 is an early career researcher with a dual background in anthropology and information technology, and has also applied the field theory method in their own community-based co-design research [50]. Author 3 is an early career researcher with a background in computer science. Author 4 is a PhD student with a background in social work. Authors 2, 3, and 4 have worked hands on as tutors with many of our students. Author 5 is a senior

academic in HCI well known for her work in engineering design and participatory design, who originally conceived the field theory method. We all share a commitment to constructivist learning, where students construct knowledge through learning activities that are hands-on, that reflect real-world design practice, and that allow students to build on what they already understand [4]. Field theories developed as a combination of our own qualitative,

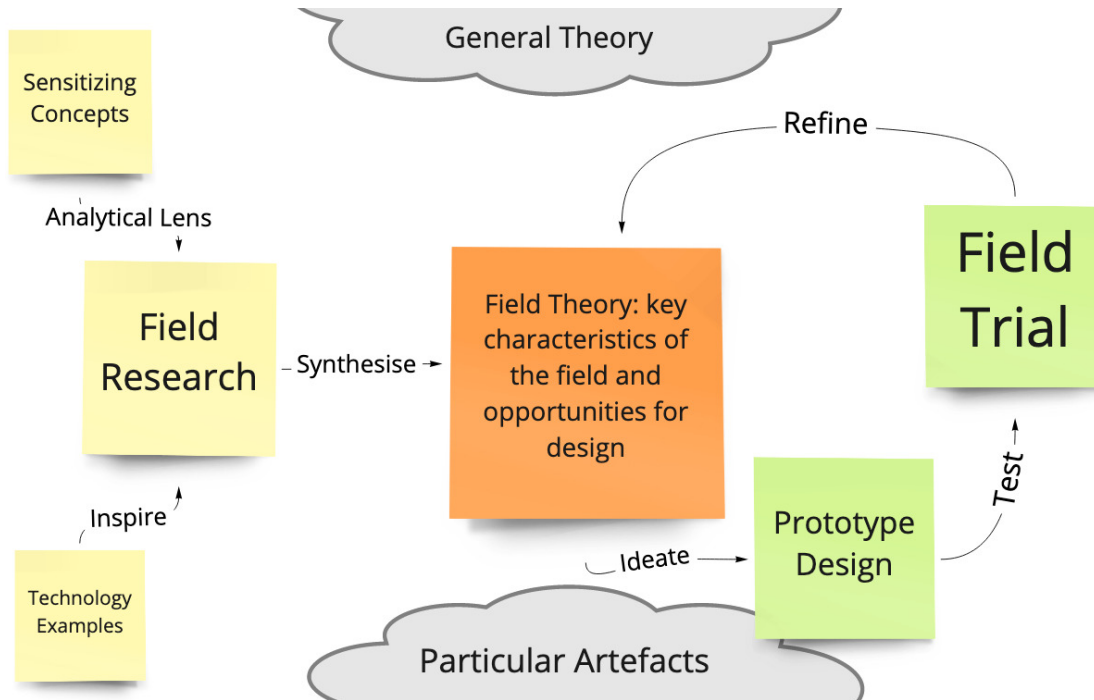


Figure 2: Students construct field theories iteratively: (1) an initial field theory is constructed from discoveries from field research, sensitizing concepts that frame research and data analysis, and inspiration from demo sessions with prototypes developed by our research team. (2) Students refine the field theory through designing a prototype and evaluating the prototype in the field.

participatory, and design-led approaches, our commitment to constructivist learning, and our desire to help students develop a working knowledge across different design approaches.

3.3 Designing with Field Theories in the Classroom

Field theories are constructed iteratively through a series of design research activities. Figure 2 provides an overview of the different activities and the order in which we teach them. From left to right, the diagram shows that students create an initial field theory through field research, followed by designing a prototype and evaluating it in the field to test and refine the field theory. The illustration is also inspired by diagrams of intermediate-level knowledge [30], which occupy the space between particular design artefacts (on bottom) and broader theories (on top). Hence, before delving into field research, students interact with technology examples from our research lab, and we also introduce students to HCI theories to help them frame their research. These activities expose students to both abstract and particular knowledge, which they need to synthesise through their field theory.

The activities described above and in Figure 2 are flexible, and over the years we have tried different types of field research, prototyping, etc. The specifics of these activities described below are based on last course in 2020, which started face-to-face and then changed to online learning due to COVID-19 pandemic. In 2020, field theories were marked through 3 assignments: assignment 1 (worth 35% of the overall mark) was presented as a report

to describe the field research, sensitizing concepts, technology examples, and the initial field theory (diagram and a textual description). Assignment 2 (worth 40%) was a group presentation to demonstrate the prototype design and its relationship to the field theory that inspired it. Assignment 3 (worth 25%) was an individual assignment report of the field trial and refined field theory.

Sensitizing Concepts: we start by introducing students to theories from HCI and related fields that offer background ideas for their research (also known as sensitizing concepts [8]). In week 1 students learn about the threads of experience [33] to frame their work around the notion of experience. Once field research is under way, we provide a short introduction to theories that might alert students to important aspects of their field of inquiry (e.g., illness work [12]) or their design (e.g., affordances [37]).

Example technologies: we bring several emerging technology prototypes to the workshop that have been designed and studied by our lab (IoT, social robots, wearables). Students interact with these to reflect on interactional concerns and to identify design opportunities.

Field research: we provide students with broad topics based on our own interests (e.g., enhance the experience with health services). Students conduct interviews, participant observation, and cultural probes [24] with a small group of participants. Based on a thematic analysis [9], students learn how to generate abstract knowledge (codes and themes) from their data.

Synthesising the field theory diagram: The field theory is constructed iteratively from insights from these activities to illustrate

key characteristics of the field and to highlight design opportunities. We start by sketching mind maps that describe key themes, interview quotes, surprising observations, and sensitizing concepts that resonate with observations. The process of sketching encourages a pragmatic pruning of insights. Instead of presenting everything they have learnt, students need to decide what is most interesting about the data and how they prioritize and illustrate insights. Field theories do not prescribe a specific visual structure, because any structure that we propose would also prioritize a certain kind of insight. Instead, we want each student to develop a structure that they see fit to communicate insight and design opportunities. To support students in finding an appropriate visual structure, we provide examples of various diagrams and we also invite students try different visual structures, e.g., timeline, mind map, storyboard. Edward Tufte's books, such as the *Visual Display of Quantitative Information* [51] and *Visual Explanations* [52] also provided inspiration. A brief title and tag line are added to the field theory diagram to guide the reader. The ultimate aim is for the diagram to work as a standalone representation of key insights, as illustrated in Figure 1. It also serves as a basis for discussion with the teaching team who might highlight missing aspects, such as how exercises might fit into daily routines, be located, be made social, etc.

Prototype design: this step is a group activity. Students discuss each other's field theories and select the most promising theory to generate ideas. The selected theory provides a set of requirements, through which students identify potential touchpoints and generate at least 30 design ideas. These ideas get refined through peer feedback. One of the ideas is developed as a prototype. Due to COVID-19 restrictions, students created video prototypes [32] that could be trialled over a distance.

Field trial: students design a study of their prototype with at least 2 participants to learn about their experience, with the ultimate aim of testing, clarifying, or providing further insight into their field theory. Based on the findings, field theories get refined, extended, revised, or at times disbanded and created from scratch. Again, the emphasis is on sketching a diagram and refining the theory. This encourages students not to over hype the presentation of their design idea in a marketing pitch, glossing over real use. Instead it encourages them to reflect on how they can refine the theory (as grades reward this!). A written report provides background about the design process and empirical evidence.

4 STUDYING FIELD THEORIES

Our aim in this paper is to reflect on potential strengths and challenges for learning and teaching interaction design research through field theories. Our reflection is based on interviews with former students, student assignments, and our own personal observations. This study received ethics clearance from our university.

We conducted semi-structured interviews with 14 former students in our interaction design course. All students were enrolled in undergraduate degrees in computer science or information systems - none of them had any prior training in design. The students who volunteered for our interviews were largely students who did well in the course (12 out of the 14 participants received a high distinction of 85% or above). However, these students worked on different topics and had different experiences with field theories because they

studied in different years. Seven students (P1-P7) were interviewed face-to-face in mid-2017, after students had completed their course. A further seven students (P8-P14) were interviewed in mid-2020 via video-conferencing. These students had their course disrupted by COVID-19 midway through the semester. During the interviews we went through the student's assignment to discuss their field theory and their learning experience. We asked students to talk us through the different steps taken to create and refine their field theory, and to reflect on challenges and lessons learned. Interviews lasted between 30 and 60 minutes and were audio-recorded.

To gain a better understanding of the learning experiences of all students, including students with lower marks, students in our 2020 course had to write a reflection on their field theory as part of their final assignment, which we incorporated into our data analysis. Students were asked to reflect on their personal experience in designing with field theories, lessons learned, impact on the prototype design, challenges experienced, and recommendations for improving the course offering. We acknowledge that these reflections are limited, because they were marked and hence students may have been more positive in their reflections.

Finally, we included our own observations and personal notes from teaching field theories over the last 5 years in our analysis. These data included personal notes kept electronically by the first author: weekly meeting notes, feedback given on assignments, reflections on assignment marking process, student feedback provided through university-administered surveys and student emails, and our responses to such feedback, and our rationale for changes to the course design over the years.

Our analysis was inductive, iterative, and collaborative. The first author immersed himself into the data by listening to interview recordings, followed by analysing interview transcripts and assignment reflections in NVivo to establish common themes in the learning outcomes and challenges throughout the process of synthesising and testing a field theory. Author 1 also reflected on observations throughout the last 5 years of teaching by going through assignment instructions and feedback given to students and generated the initial paper draft. All authors discussed and edited the draft, with author 2 also bringing in their personal experience in constructing a field theory for their thesis to refine the definition of field theories and the discussion.

5 FINDINGS

5.1 Sensitizing Concepts: Grasping Relevant Ideas

A first step for creating field theories is to learn about existing theories and to identify sensitizing concepts that frame the research or highlight important ideas. This was a new and challenging activity for many students. They commented on the difficulty of grasping the ideas and the relevance of HCI theories for their project: *"this was pretty challenging as some of the existing theories could have a heavy language . . . and it was very difficult to connect it to our own case."* (P6)

According to students, HCI theories that provide a concept that is intuitively graspable have been most helpful. Theories of motivation and behaviour change [19] were particularly popular with students, because they were easy to understand and they also

provide a lens that highlighted important aspects of the findings (i.e., motivations and related challenges) for the field theory: *“For example, Self-Determination Theory gave me some insight into why cyclists feel may feel more motivated during a ride than beforehand (because they are experiencing competency), which I could then ask participants about to confirm if this was their experience.”* (Assignment) Similarly, the field theory in Figure 1 highlighted motivational challenges, like boredom from repetitive exercises and lack of visible progress. Students also valued the concept of affordances [37], which is easily graspable and provided them a new perspective on designing interactions. However, the assignments often only showed a partial or superficial understanding of these theories, and motivational concepts rarely inspired strong design ideas.

Based on our observations, theoretical concepts need to be graspable but also provide students with a layered framework to help them unpack specifics of their research. The threads of experience framework [33] has been useful, because it helped students to unpack how our senses and our perceptions of space and time contribute to an experience. Some students found the framework also useful to evaluate experiences in their field trial: *“Designing interactions was especially helpful with the ‘threads of experience’ content. This content allowed me to think of interactions differently, allowing me to think deeper and better evaluate experiences.”* (Assignment). Hence, in the last two years we made this framework mandatory for all students.

5.2 Technology Examples: Inspiration from Prototypes

Technology examples expose students to current trends in interaction design, similar to how sensitizing concepts expose students to current theories. In week 2 we introduce students to technology prototypes (IoT prototypes, social robots, and wearables) designed or studied in our lab. Many students commented that interacting with these technologies was one of their highlights, and that it changed their view on design.

“One thing that I loved about the unit was those emerging technologies. How the robot and then the emoji app and you know all that stuff really interacting with them and being able to see them in person and actually look at them and interact with them. . . . It really changes your whole perspective on it, because you can read about it and look at all the theories and robots and machine learning and IoT and stuff like that. But really seeing it in person makes a big difference.” (P11)

Based on the threads of experience [33], students write a reflection and create a diagram about their interaction. This helps students identify design opportunities. It also gets students started at making connections between concrete technology interactions and abstract concepts for their field theory.

5.3 Field Research: Grappling with Qualitative Data and COVID-19

Field research is essential to identify insights into relevant characteristics of the field and design opportunities for the initial field theory.

According to their assignments, students enjoyed interviews, observations, and cultural probes. Half of the students (16/31) reflected in their assignment that data collection and analysis were key skills that they had learnt.

“The lessons learnt from designing interactions and experiences with field theory and throughout these assignments are that data collection is number one, without solid and thorough data to go off, the whole process is in trouble of failing. Also, that data analysis is also key, that being able to look at raw data and convert it into information that can be worked with is critical.” (Assignment)

Students commented on the challenge of qualitative data analysis. We introduced students to thematic analysis [9] to code the data and to identify salient themes. While abstract themes are helpful to organize the data, we recognize that themes in and of themselves are not particularly helpful to convey design opportunities. In our experience, designs are often inspired from knowing the specifics of an experience and its context. Hence, students need to learn to keep connections between themes and specific experiences observed in their research. Furthermore, the field theory is not meant to be a summary of all insights from the field. Students need to select findings that present interesting design opportunities, which is challenging.

“The biggest challenge faced during the run of this semester was the actual analysis of data. . . . trying to actually make sense of the gathered information and tying it all together into a coherent mass was challenging by virtue of just how much data there is to sort through at the end of the data collection stage. . . . Some of the data would also end up being classified as non-relevant, only to be reclassified as actually relevant later on, when other bits of information were analysed.” (Assignment)

COVID-19 broke out in our city in the middle of the field research. Hence, we changed our plans to conduct field research safely over a distance. Interviews were conducted via video conferencing tools and probe activities were designed so that participants could complete them via their mobile phone, note tools, and online whiteboards. Instead of participant observation, students relied on diaries and photos taken by participants to indirectly observe experiences in daily life. Some students alluded to how COVID-19 affected them personally, through concerns about their studies, social isolation, mental health, and concerns about their future. However, P10 also stated that the field research provided him with a legitimate reason to connect with others during the lock-down: *“Talking to participants was interesting. I got to talk to my friends, especially during COVID-19. It’s an opportunity to talk to a person you like.”*

5.4 Drawing a Field Theory Diagram: Defining a Scope and Structure

Qualitative data analysis and drawing the field theory diagram usually go hand in hand. The thematic analysis helps to identify salient themes and provides an initial structure for the diagram (e.g.,

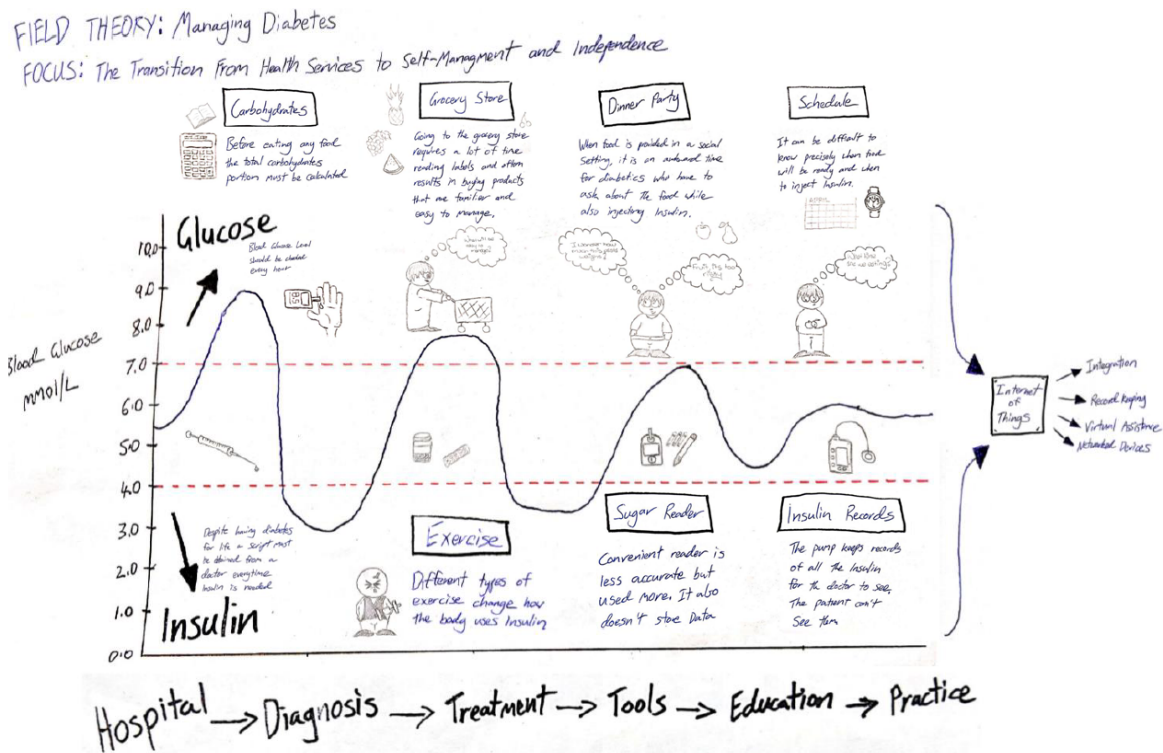


Figure 3: Field theory of the ups and downs of living with type 1 diabetes. Structured around the concern of managing blood glucose levels, the diagram presents concrete experiences of counting carbohydrates, checking labels during grocery shopping, and the awkwardness of managing insulin when invited to a dinner party with little control over food.

when presented as a hierarchical map): “the coding structure and then a thematic analysis on top of that, that really brought all of my ideas together and it really structured to those so that I could separate it into the separate parts of the field theory.” (P11) Conversely, drawing a diagram helps to analyse data because it forces students to select relevant data and to make relations between different themes and concrete experiences explicit.

Creating the diagram was often the first thing students mentioned during interviews. Students recognized that diagrams are powerful to analyse data from different perspectives, and to communicate insight. However, they found it challenging to draw them, because at that stage in their projects it was not clear what a field theory is and what its value may be: “You got to build the field theory diagram without knowing what it was.” (P14).

The challenge is about scope (what findings to include) as well as about the narrative (how to tell a story that integrates the various findings, presents a clear argument, and also offers a rich understanding of experiences). Students particularly commented on the lack of pre-defined structure for their field theory diagram: “Step 4 [drawing a field theory diagram] was challenging because we didn’t have any ‘recipe’ on how to do it.” (P7) We do not give students a template to fill in, because that would prioritize a certain kind of insight. Instead, we ask students to create the diagram in a way that they see fit to communicate key characteristics of the field and design opportunities, and we support them by providing diagram

examples and by encouraging them to try out different structures (a mindmap, a network, a journey) which we critique with them.

From the assignments and interviews we identified common tactics. Students typically imitate the structure of an example that we present to them, like the journey depicted in the rehabilitation field theory in Figure 1. Many students lay out their key themes, and if space permits, add specific examples and quotes to illustrate themes. Some students use theories as a lens to filter what themes they include or exclude from their diagram, e.g., by laying out the themes from self-determination theory (competence, autonomy, relatedness). A few students produced excellent results by identifying a central idea or challenge, which ties together several of their findings. For example, the diagram presented in Figure 3 is structured around the challenge of managing blood glucose levels, which is central to the lives of people living with type 1 diabetes. Having to manage blood glucose levels affects not only what they eat, but also how they shop for groceries, work, exercise, etc.

5.5 Field Theory: Qualities of Working Insights for Design

We have marked approximately 180 field theories during the last 5 years. When we mark the first iteration of a working field theory, we assess the field theory diagram and description based on the following qualities: (1) integration: the theory integrates insights from field research with theory and technology examples (which

are presented separately in the assignment report); (2) clarity: the diagram is well structured and easy to comprehend; (3) credibility: the insights are authentic and based on empirical data, (4) relevance: the theory highlights clear opportunities for design. Most students are able to create a field theory that meets these qualities and pass the first assignment.

In our open-ended feedback to student assignments, three qualities have stood out over the years that make a field theory highly relevant for design. The first quality is experience over abstraction. Strong assignments highlight experiences through quotes and anecdotes from the field, including undesirable and desirable experiences. These field theories illustrate the experience from the perspective of participants, e.g., in a first person-view of a person with an injury (Figure 1) or a person living with type 1 diabetes (Figure 3), rather than purely through abstract themes and concepts. They tend to indicate that the student has taken the fieldwork seriously rather than simply construing ideas from their own preconceptions.

The second quality is originality from specifics. This is where the field theory rings true and also presents something new that goes beyond the obvious and shows analytic insight. Originality often comes from presenting the specifics of an experience, e.g., Figure 3 illustrates the challenges for a person with type 1 diabetes to manage insulin when invited for dinner parties due to the lack of control over the timing and ingredients of food served. The threads of experience framework [33] was helpful for students to unpack details of an experience. These specifics can reveal interests or gaps to investigate further.

Finally, outstanding field theories present a central concern that integrates empirical data, theory and design. Instead of presenting themes in the data and concepts side by side, these theories synthesize the information and present a clear narrative and scope around a central concern. In figure 1, the integration is well done by highlighting different motivational challenges. Figure 3 is structured around the challenge of managing blood glucose levels. The diagram uses the metaphor of a blood glucose charts to show ups and downs in living with type 1 diabetes across different experiences, e.g., during shopping and preparing for meals, exercising, and at dinner parties and other social events where the ingredients and the timing of a meal cannot be controlled.

5.6 Prototype Design: A Response to the Field Theory and a Means to Test the Theory

A main strength of a field theory is its role in supporting the development prototype designs. More than half of the students (18/31) commented in their assignments that the field theory provided a clear scope for their ideation and prototyping by displaying relevant characteristics of the field and design opportunities. At this stage students start to recognize the value of a field theory, and some students stated that they would not have been able to come up with their prototype design without their field theory.

“I really enjoyed just the design of the prototype. Because it was, it was kind of that tangible reward or the or the combination of the research. For me I just really found that to be the most fulfilling thing is that coming up with ideas and then executing those on. But I don’t think that would have been possible without

the field theory. As you know, a strong visualization of what the problem spaces actually were and communicating that with the team to generate those design ideas.” (P8, see also Figure 4)

The prototype design is also a means to test the field theory and refine it through new insights, which from a teaching perspective, has several benefits. We have seen a more reflective approach to how students present their designs. The assignment emphasises insights into field theories as the main outcome. Hence, instead of pitching the perceived novelty of their prototype design, which tends to propel students into an uncritical and overly effusive marketing mode, students need to reflect on how their design evolved from the initial field theory and how it helped them gain further insight to refine their field theory: *“the prototype helped to create questions to ask participants - it was really an excuse for finding meaningful data.”* (P5)

Field theories have also been useful to orient design ideas towards specific experiences (and away from trying to solve a broad, underspecified problem). For example, for topics focussing on food and cooking, in our first two years of teaching we have seen an abundance of ‘smart fridge’ and ‘smart kitchen’ ideas that automate tasks and make the human redundant. However, by emphasising actual experiences as requirements in the field theory, we have been able to guide students towards designs that enhance a person’s experience and provide insight for the field theory. For some students this has also been useful for their prototypes to consider how they can implement their ideas in a way that creates a new kind of experience, e.g., through a Wizard of Oz approach. The most interesting designs have applied this to create experiences that connect people in new ways, as illustrated in Figure 5

We have found it difficult to strike a balance between the importance of the prototype design and the field theory. In the first few years we did not see many interesting prototype designs. In hindsight, we suspect that by emphasizing the field theory as the main outcome of the assignment, we may have inadvertently given students the impressions that the prototype design is not important, as many prototypes were poorly implemented or merely a sketch. Many other designs were fairly generic implementations of social media or gamified designs, with little consideration to the specifics of their field theory. Hence, in 2020 we made a number of changes which led to innovative and mature prototypes, without diminishing the quality of the final field theory. We re-designed the assignment to give more weight to the prototype. We also made it a group assignment so students can select a field theory with interesting gaps and design opportunities from amongst their team members, and collaborate on developing ideas and prototypes. Due to the COVID-19 lockdown, we also changed from physical prototypes to video prototypes. Students had to create a video that introduces an opportunity highlighted by their field theory. They also needed to roleplay and show how a person interacts with a prototype in a specific context. This helped students to create designs that were specific to their field theory and design situation, as in the GrowBot design (see figure 4).

5.7 Field Trial: Refine the Field Theory

The final step in the assignment is a short field trial of the prototype design to refine the insights from the field theory. Participants are



Figure 4: The GrowBot video prototype design by P8 was aimed at people living in apartments who want to learn how to grow food. Inspired by practical challenges like finding a suitable location and providing enough water, the GrowBot design explores how life-like expressions can guide growers and cultivate knowledge and care for plants.

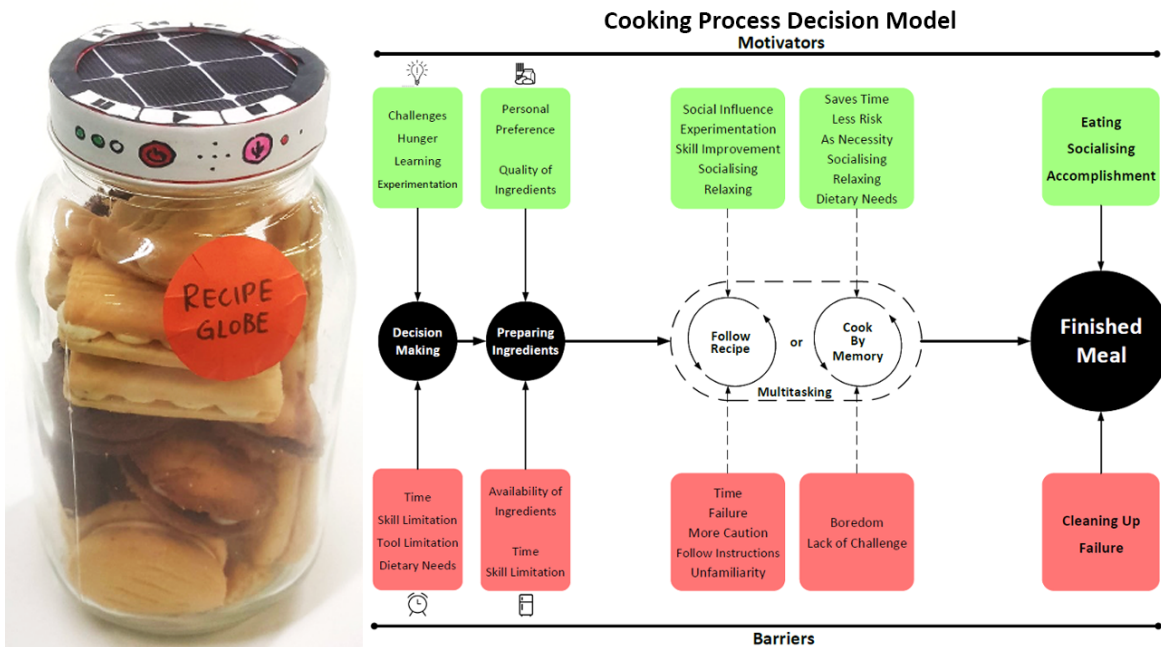


Figure 5: Recipe Globe design and original cooking field theory that inspired the design. The design seeks to enhance the cooking experience by allowing people to record and share recipes as well as the goods that they have baked. The student (P6) used a Wizard of Oz approach to simulate voice interaction with the Recipe Globe by controlling a voice recorder in the kitchen, which also allowed her to observe the participants' interactions and experiences. One participant in her study recorded a recipe during baking, and a second participant tried to follow the audio-recorded recipe to make cookies herself.

invited to view a video prototype or to interact with a physical prototype, followed by an interview about their (imagined) experience with the prototype design. Even with only two participants, students gain valuable insights: *"There were a couple of things that I didn't expect to be mentioned or to be pointed out, and I went, ah, those are some pain points."* (P14)

The trial is an essential step of the field theory because it brings the design research process back to the field theory. We emphasise to students that the purpose of the trial is not to find out whether

people like the design. The purpose is to see what the design would lead to, how it might change experiences, and what the design reveals about the field theory that inspired it. This helps to prevent students from pitching their perceived novelty of their design. Instead, they focus on refining the field theory diagram for their final report.

"Another challenge was how easy it was to get wrapped up in the excitement of designing something, that it was easy to lose sight of the role of the prototype as a tool for primarily exploring the field theory. I

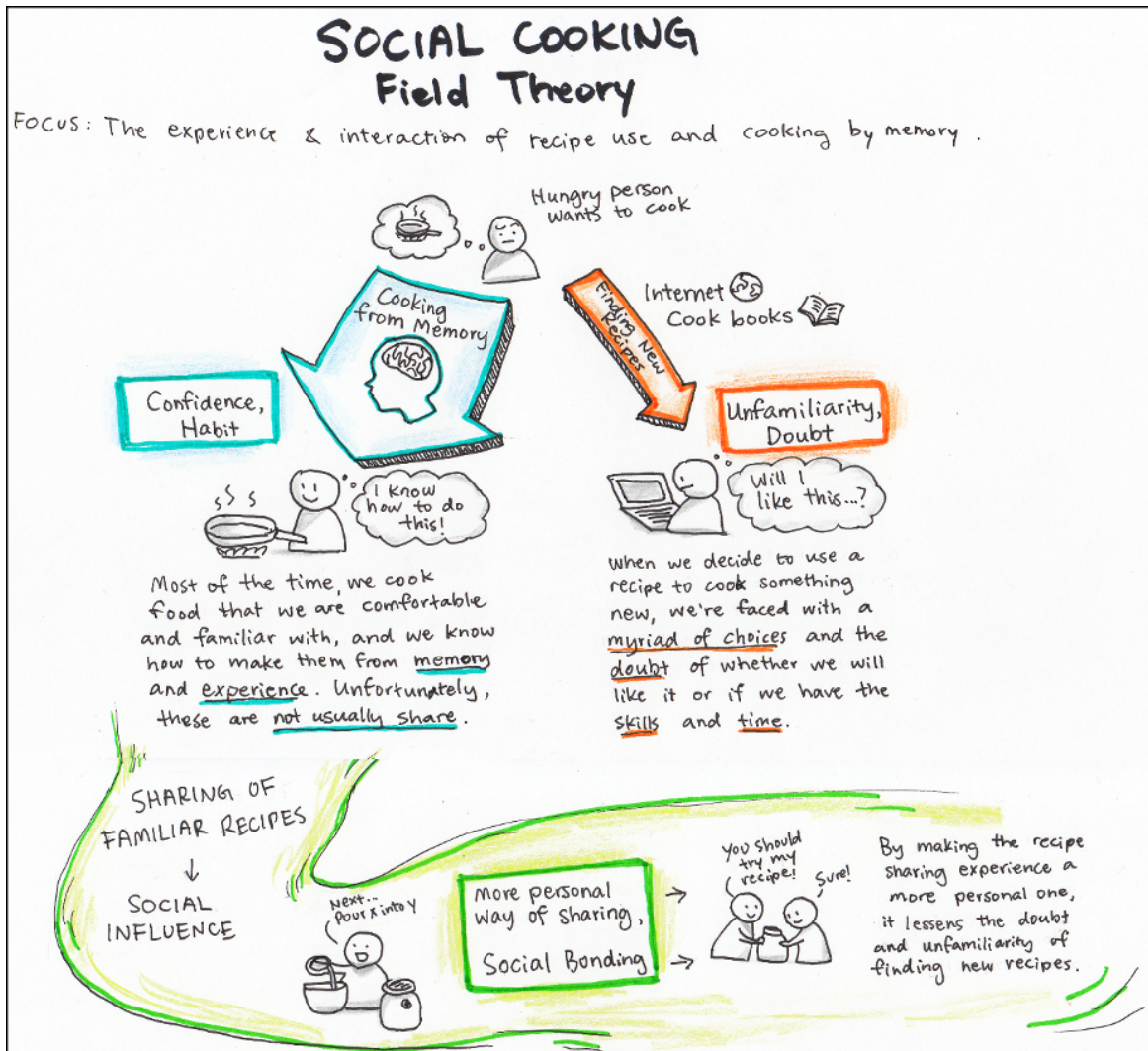


Figure 6: Refined cooking field theory. The findings from trialling the Recipe Globe design (depicted in figure 5) led to a completely revised field theory, which highlights experiences with cooking from memory (teal), cookbooks (orange), and shared recipes (green).

kept on having to re-centre and remind myself of what the goal of the research actually was – to understand people and experiences, not just design something cool.” (Assignment)

Most field theory refinements are minor. Students typically expand on existing sections of their field theory based on their trial study findings. Only very few students make major revisions to their field theory. For example, Figure 6 shows a largely revised cooking field theory based on trialling the Recipe Globe (Figure 5) with two participants. One participant recorded a recipe with the Recipe Globe, and a second participant recreated the meal in their own kitchen. Like the original field theory (Figure 5), the revised field theory in Figure 6 centres around cooking from memory and cooking from recipes. However, P6 removed all other parts from

the original field theory and instead the final field theory presents a more nuanced understanding of the experiences with cooking from memory and from recipes, as facilitated by the Recipe Globe design.

6 DISCUSSION

In this paper we have presented working field theories, which integrate insights from fieldwork more tightly with design research. Field theories occupy a unique position amongst design approaches in HCI. They visualise fieldwork insights for design, similar to experience models [6], but field theories also get evaluated and refined further through design research. In this regard, field theories resemble a form intermediate-level knowledge [30] with new insights generated from artefact designs and use studies. Unlike other forms of intermediate-level knowledge, however, field theories communicate the very insights and assumptions that have led to a design

in the first place, and that allow others to generate new designs. In other words, field theories make the connections between fieldwork and artefact design more explicit. Students learn to move between field research and design activities, and to co-evolve their field theory and prototype design in parallel.

Drawing a field theory diagram is a key feature of our approach, because it requires that students synthesise insights for design from fieldwork, technology examples, and sensitizing concepts. Students explore different visual structures to elaborate on their insights, and they also need to make decisions on what empirical insights and sensitizing concepts to include. Such visual thinking [2] is common to students trained in creative design [54], but it is often a new skill that computer science students develop when they create a field theory. Visual thinking skills are critical to the design process more broadly, to communicate design ideas and to make decisions on the design of products and services [10; 26]. The diagram is also important to discuss insights with students, similar to how designs get critiqued in studio-based approaches to design [53].

Testing the field theory is important to connect design outcomes back to initial insights from field research. Here, the field theory integrates two forms of design knowledge in a unique way. On the one hand, a field theory is a working theory *for* design that frames the inquiry and highlights design opportunities [55]. On the other hand, the field theory is also a form of knowledge that is refined further *through* design by building and trialling a prototype [56]. Striking a balance between the importance of the prototype design and the field theory is challenging for computer science students. However, we have seen that field theories encourage a reflective approach to presenting the novelty and significance of prototypes. It also focuses the scope of field trials to inquire further into the concerns and desires expressed in the initial field theory. The focus on the problem-opportunity space, rather than the design outcome, and the feedback from trial studies to refine the problem-opportunity space, set field theories apart from research-through-design [23; 25; 30; 56].

6.1 Implications for Learning and Teaching HCI

We hope that field theories will help HCI educators and students to bridge different ways of learning and teaching in computer science and design schools. For HCI educators in computer science, field theories offer a way of incorporating creative design principles of ideation, visual thinking, divergent thinking, and peer critique [53; 54], with the iterative design approaches typically taught in computer science [54]. In making field theories central to the design process, educators can bring HCI theories to the process, and beyond that, students can synthesize their own working theories based on fieldwork. Diagramming a field theory provides a clear scope and direction for the ideation and prototype design activities, which can otherwise be daunting for students not trained in ideation and divergent thinking. In framing the prototype design as a means to test the field theory, students learn to be critical about their design and more attuned to the specifics of a design situation.

HCI educators in design schools may also find field theories useful for their teaching to open up the 'black box' of creative design [18]. Instead of reflecting on and abstracting from design

instances post-hoc, field theories provide a flexible format that students can incorporate into their portfolios to articulate relevant ideas, empirical insights, theories and personal preferences. Field theories keep the problem-opportunity space visible throughout the design process and allow for peers and educators to interrogate and critique them.

For students interested in learning designing with field theories, we highlight that the approach presented in this paper is not meant to be a recipe with step-by-step instructions for producing a practical theory. Instead, we have found Levi-Strauss' notion of a 'bricoleur' [31] to be a useful analogy. A bricoleur is "a Jack of all trades, or a kind of professional do-it-yourself person" [15, p.2]. Similarly, our aim with the field theory has been to equip students with skills to employ design methods and theories to address the task at hand. Unlike a traditional HCI researcher, a bricoleur does not necessarily go into depth of a theory, or ground the field theory by making connections to a wider set of theories. Instead the bricoleur is familiar enough with theory and practice across computer science and design; someone who can move comfortably between these fields, look at different bits of a theory, empirical data, and design ideas, and re-purpose them to deliver practical outcomes.

The notion of a 'bricolage' is also a useful analogy for HCI educators interested in teaching design through field theories. Field theories are a flexible method that HCI educators can 'cobble together' with their own research problems and their own pedagogical approaches. The creation of field theories through diagramming is central to synthesize insights from empirical research (including technology trials in the field), and to incorporate theoretical concepts. However, HCI educators can apply field theories to their preferred research methods, HCI theories, and technologies. In terms of methods, we have focused on experience-centered design [33], but we can also see how field theories could work well with design methods that foreground values [21], participation [47], or entanglements of humans and technology [20].

6.2 Future Work Towards Academic Field Theories

In revising field theories over the years, we have also reflected on how to balance between the needs of the HCI classroom and the rigor expected from academic theories. To further enhance the rigor of field theories, e.g., for graduate students wanting to incorporate field theories into their research and into peer-reviewed publications, we can see how field theories could benefit from the steps of horizontal and vertical grounding that are part of the process of constructing a strong concept [30]. In examining the horizontal grounding of field theories, students would examine how their insights and design opportunities apply to other contexts, e.g., by reviewing related work or by exploring their insights and designs in a related context. In vertical grounding, students might explore how their field theory relates to other theories as well as to other technology designs. To further enhance the credibility, field theories would benefit from deepening the relations with participants, e.g., by checking the theory with the members that it seeks to represent, or through a participatory approach where participants and researchers co-construct the theory [5].

For graduate students, a field theory can be useful to develop a scope from initial field research and concepts, and to document how central concerns and insights evolve throughout different research and design activities of their project. The flexible format of field theories can also be helpful to synthesize insights from diverse methods and data, where traditional qualitative research methods like grounded theory [11] may not be appropriate. However, more research is needed to explore how well field theories scale up to projects that involve a large number of participants, run over multiple years, and lead to multiple iterations of a field theory. To manage the large set of data, graduate students need to be more selective, and reflective, about what findings they include in their field theory. They may also need to condense and abstract their empirical insights to a larger degree in order to enhance the clarity of the field theory. However, based on our findings, we caution against entirely abstract field theories. Whilst contextual information can be abstracted, field theories that include concrete experiences and specific quotes tend to present a more interesting path for further design and research.

7 CONCLUSION

Achieving mastery across fieldwork, theory and technology design is certainly an art that requires practice and experience. The field theories presented in this paper offer a starting point for students with a technical background to develop a sensibility for fieldwork, theory and design, and to integrate these skills by diagramming and trialling field theories. In laying out the steps for constructing field theories together with our assumptions and feedback from students, we offer an approach that HCI educators and students can integrate into their own classrooms and design projects. The approach presented in this paper focussed on working theories. However, we have also laid out a path forward for enhancing the rigour of field theories so that interaction design researchers can create field theories that offer original and significant academic contributions.

ACKNOWLEDGMENTS

We thank all the students in our classes, our colleagues in the QUT CHI Writing Workshops, as well as the anonymous reviewers for their support.

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